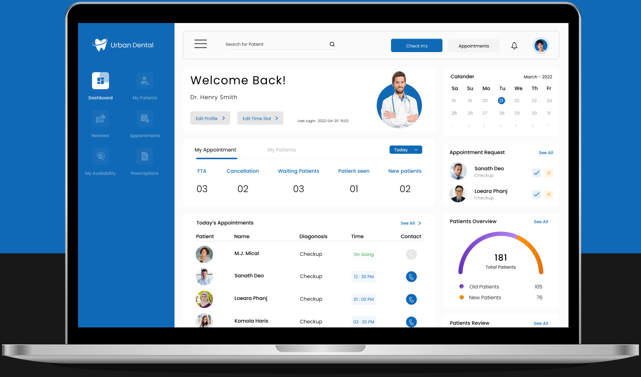
URBAN DENTAL

Case Study



Urban Dental
Dashboard
My Patients
Reviews
Appointments
My Availability
Prescriptions
Search for Patient
Check In's
Appointments
Edit Profile
Edit Time Slot
Calander
March - 2022
Sa Su Mo Tu We Th Fr
18 19 20 21 22 23 24
25 26 27 28 29 30 31
1 2 3 4 5 6 7
My Appointment
My Patients
Today
Appointment Request
See All
Patients Overview
See All
181
Total Patients
Patients Review
See All

Welcome Back!

Dr. Henry Smith

Last Login: 2022-04-20 15:02

FTA	Cancellation	Waiting Patients	Patient seen	New patients
03	02	03	01	02

Appointment Request

Sanath Deo	Checkup
Loeara Phanj	Checkup

Today's Appointments

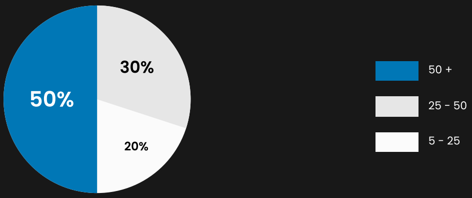
Patient	Name	Diagonosis	Time	Contact
	M.J. Mical	Checkup	On Going	
	Sanath Deo	Checkup	12 : 30 PM	
	Loeara Phanj	Checkup	01 : 00 PM	
	Komola Haris	Checkup	02 : 30 PM	

Old Patients	105
New Patients	76

Overview

Urban Dental is a clinical management tool that leverages the power of digital to offer a convenient, customizable, completely functional solution to operate smoothly dental clinics and their dentists.

By the research, clear idea of the demographics is estimated that 50% of the patients are 50 years old and over, children represent about 20%, the rest (30%) are 25 to early 50's years of age.



Demographics chart of patients visiting dental clinics

Age Group (Years)

The Problem

Running a dental clinic means juggling appointments, patient records and payments, much of it on paper. For patients, booking a visit takes time and energy when they are already in pain.

Solution

Create a tool that gives dentists their appointments, availability and patient records in one place, and lets patients book a visit without having to call and receive timely reminders.

Some Common Problem Statement

Not everybody knows who we are, they just don’t believe we are allowed to do this under professional supervision.
- Dental Co-Assistant

It’ll be easier if we have the patient profile, their complaint, or proof-like photo of their mouth.
- Dental Co-Assistant

The worst thing when you don’t have any patient to check on, you literally need to walk to a neighborhood and knock on their door.
- Dental Co-Assistant

It would be nice if I know the complete professional profile of the Co-assistant, I don’t want to be fooled.
- Patient

It’s hard to suit my schedule with theirs.
- Patient

I don’t know much about the hospital and their program, maybe if you could give us more information about these kids that want to treated toothache for free it should draw more attention.
- Patient

Design Process

Discover

Define

Ideate

Design

Test

Discover

Executive Intent:

This project is about making it easy for users to book a dentist appointment

Target Audience

Gender: Male, Female and Transgender
Profession: Working or Non - Working
Age group : 18+

Cross Platform

Web app

Define

Competitive Analysis

At this step, understand the services and features provided by different competitors around the globe. In the first phase, I did secondary research, such as looking for each websites and their users reviews. I found the information, motivation, and problems that users often experience while using these services.

List of Competitor

Competitor List	Overview	Active Since	Services
curve DENTAL Website	Curve SuperHero practice management software makes everything easy. It's goal is to deliver cloud-based software that makes it easy to manage a dental practice.	2004	• Implementation • Data Conversion • Data Management • Training • Support
practice-web Helping Dentists Thrive! Website	Practice-Web is a comprehensive practice management solution designed for today's modern dentist.	1988	• Comprehensive Practice Management • Tutorials and Webinars • Free Demo • Training & Support
DENTRIX ASCEND Website	Dentrix Ascend includes a predictive problem-solving feature called Goals, Problems and Solutions (GPS). This unique business tool identifies tasks that need attention and provides solutions to keep your practice on course.	2014	• Multisites • Practice Management • Front Desk • Clinical
dovetail .co Website	Dovetail is the world's only truly mobile, easy to use Dental Practice management software that will actually save you money. Dovetail offers a mobile EHR web-application that is certified by the ONC and is HIPAA compliant.	2010	• Front desk & Practice Management • Clinical & EHR • Patient Engagement • Business Management & Analytics
planet DDS Website	Denticon was the first cloud-based dental office management software. It is a reliable program suitable for practices of all sizes. Denticon dashboards are designed to be intuitive so smaller teams can achieve more through easy user interfacing and useful reports.	2003	• Modern Clinical Experience • Easy-to-use Scheduling • Efficient Back & Front Office Operations • Online Patient Tools • Insightful Analytics • Native Patient Communications • Multi-Specialty Capabilities • Enterprise Management • Advanced Security

Feature comparisons

Comparison of functionality within competitors

Feature	Curve	Practice-Web	Dentrix Ascend	Dovetail.co	Denticon
Imaging	🟢	🟢	🟢	🟢	🟢
Charting	🟢	🟢	🟢	🟢	🟢
Progress Notes	🔴	🔴	🔴	🟢	🔴
Treatment Plan	🔴	🟢	🟢	🟢	🟢
Explosion Codes	🔴	🔴	🔴	🟢	🔴
File History	🟢	🟢	🟢	🟢	🟢

Medical History	🟢	🟢	🟢	🟢	🟢
Scheduling	🟢	🟢	🟢	🟢	🟢
Billing	🟢	🟢	🟢	🟢	🟢
Online Forms	🟢	🟢	🟢	🔴	🟢
Patient Portal	🟢	🟢	🟢	🟢	🟢
Custom Reports	🟢	🟢	🟢	🟢	🟢
Payment Processing	🟢	🟢	🟢	🟢	🟢
Perio Charting	🟢	🟢	🟢	🟢	🟢
E-Prescription	🟢	🟢	🟢	🟢	🟢
Quick Exam	🔴	🔴	🟢	🔴	🔴
Insurance Portal	🟢	🟢	🟢	🟢	🟢
HIPAA compliance	🟢	🟢	🟢	🟢	🟢
Automated Text Reminders	🔴	🟢	🔴	🟢	🔴
Teledentistry	🔴	🟢	🔴	🔴	🔴
Patient Texting and Reviews	🟢	🟢	🔴	🟢	🟢
Doctor Portals	🟢	🟢	🟢	🟢	🟢
Security and System Requests	🟢	🟢	🔴	🟢	🟢
Database Conversion	🔴	🟢	🔴	🔴	🔴
Patient Routing	🔴	🔴	🔴	🟢	🟢
Location Access Restrictions	🔴	🔴	🟢	🔴	🔴
Monitor Business Metrics	🔴	🔴	🟢	🟢	🟢
Add more Offices	🔴	🔴	🟢	🟢	🟢
Multisites	🔴	🟢	🟢	🟢	🟢
Medication Adherence Monitoring	🔴	🔴	🔴	🟢	🔴
Behavioral Support	🔴	🔴	🔴	🟢	🔴

SWOT Analysis

SWOT Analysis

Competitor List	Strengths	Weaknesses	Opportunities	Threats
curve DENTAL	• Cloud-based all in one practice management solution • Brilliantly designed user interface	• There is no electronic signature availability for the patient file • No option for CBCT backup	• Over 41,000 dental professionals in the United States and Canada	• Poor billing and transaction system
practice-web Helping Dentists Thrive!	• Oldest service provider in industry • They provide database conversion for clinic management	• Lack of "cloud" support. • The integration process was tricky, and a bit complex.	• It provides pay-by-text feature • It's teledentistry feature differs it from others	• Only compatible with the Microsoft.NET platform

DENTRIX ASCEND	• The cloud based service and always evolving with updates impress users as they improve the product as per recommendations	• Users find its ledger quite confusing • Its insurance setup is user-unfriendly • Inability to edit procedure codes	• Its multisites service makes the product outstanding to save money and time • Keep track of multiple clinics	• While being next generation technology and cloud based it has been very slow to publish APIs
dovetail.co	• World's only truly mobile Dental Practice management software • Less expensive than similar products	• Insurance claims are not fully operational	• offers a mobile EHR web-application that is certified by the ONC and is HIPAA compliant.	• Its interface is generally focus on mobile based and tablet based users
planet DDS	• Its customer service is excellent • Easy to navigate even with zero knowledge of the product	• Almost every thing requires an extra button to push and takes longer than the other	• Physical and virtual security layers	• Imaging and charting qualities needs to improve as compared to others

User Persona

Persona 1

Jenna Williamson

"Patient interaction is the best part of my job; the second closest part is keeping the schedule full"

Age	25
Location	London
Profile	Doctor
Status	Single
Work	Dentist

Backstory

Jenna Williamson is a young dentist in London. Recently, she started her practice in town. Unlike other mature practices with many trained staff sharing job responsibilities. Besides operating dental procedures, she needs to make decisions for her practice and also works on some other pieces of stuff like explaining financing obligations and treatment costs to patients. As a startup, her practice has to stay connected to insurance to stay competitive. She is getting tired of the paperwork gradually.

Needs

- A dentistry management software that could increase paperwork efficiency.
- A financing product that could help her to communicate cost payments plans with patients.
- Knowing the position of the practice in the market and improving its competitiveness.

Pain Points

- Spending 15+ hours on paperwork on average every week.
- Don't know how to get good reviews and improve the clinic.
- Don't know how to inform the patients about the payment and get them to pay.

Empathy Map

SAYS

- Increase the number of patients/appointments
- Get good reviews from patients
- Get paid the maximum amount and on time for each procedure.

THINKS

- I wish there was something that would make my job easier.
- I wish there was an easier way to manage patients/appointments.

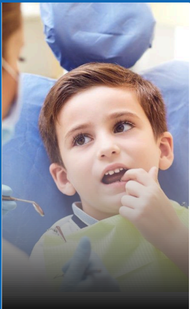
DOES

- Update the database of patients and other dentists to do job.
- Update and remind patients about the timeline and upcoming schedule.
- Keep track of payments and finance.

FEELS

- To give quality service to patients.
- To have good management platform.
- To increase the output.

Persona 2



Steve Hagerman

"I'm afraid of going to the dentist because I don't want it to hurt"

Age	4
Location	New York
Profile	Patient
Status	Child
Gender	Male

Backstory

Steve is 4 years old and has to go to the dentist because he has a toothache after eating candy. He is anxious about going because he is scared of injections, and he cries and screams whenever he gets one. Because of the toothache he can't go to school or play outside like his siblings and friends.

Needs

- Clear expectations
- To feel brave and safe
- Good dental health
- Needs an anxiety-free dental appointment

Pain Points

- Feels lied to
- Doesn't know what's going to happen at the dentist
- Avoids the dental clinic because of needles
- Doesn't feel safe but needs his toothache cured

Empathy Map

SAYS

- Doesn't want to visit the clinic
- Doesn't want the injection
- Needs the toothache cured as soon as possible

THINKS

- I wish I could be brave
- I wish I didn't get the toothache by eating so many chocolates and ice-cream
- I wish to get rewards after the dentist

DOES

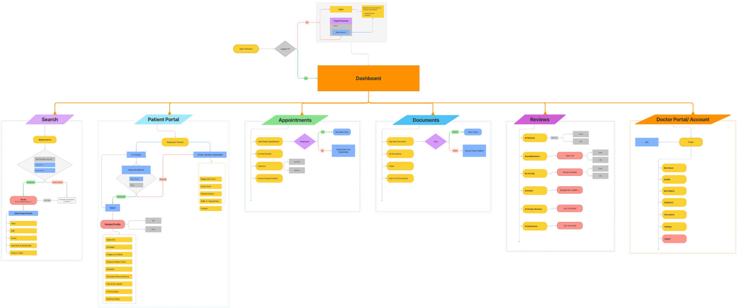
- Cries and screams when he's scared
- Asks questions about injections
- Gets scared at the dentist

FEELS

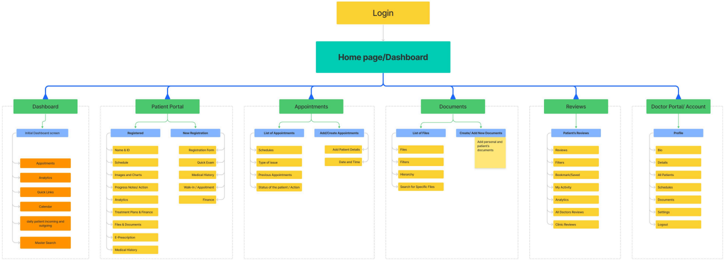
- To feel comfortable about visiting dentists
- To be fully aware and informed of what to expect
- To have good, strong teeth

Ideate

User Flow Map


Dashboard
Search
Patient Portal
Appointments
Documents
Reviews
Doctor Portal/ Account

Information Architecture


Login
Home page/Dashboard
Dashboard
Initial Dashboard screen
Appoitments
Analytics
Quick Links
Calendar
daily patient incoming and outgoing
Master Search
Patient Portal
Registered
Name & ID
Schedule
Images and Charts
Progress Notes/ Action
Analytics
Treatment Plans & Finance
Files & Documents
E-Prescription
Medical History
New Registration
Registration Form
Quick Exam
Medical History
Walk-In / Appoitment
Finance
Appointments
List of Appointments
Schedules
Type of issue
Previous Appointments
Status of the patient / Action
Add/Create Appointments
Add Patient Details
Date and Time
Documents
List of Files
Files
Filters
Hierarchy
Search for Specific Files
Create/ Add New Documents
Add personal and patient's documents
Reviews
Patient's Reviews
Reviews
Filters
Bookmark/Saved
My Activity
Analytics
All Doctors Reviews
Clinic Reviews
Doctor Portal/ Account
Profile
Bio
Details
All Patients
Schedules
Documents
Settings
Logout

Style Guide & Design

Colors

Mix of blue and white is associated with trust, knowledge, professionalism, cleanliness, calm and focus.
These qualities suit a medical product.

Main Theme	Body color 1	Body color 2	Optional		
🟦	⬜	⬜	🟥	🟩	⬛
#864631					

Typography

Geometric sans serif typeface gives a breath of freshness, modernity and gives the impression of being user-friendly.

Poppins

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z

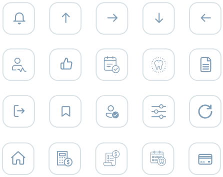
a b c d e f g h i j k l m n o p q r s t u v w x y z

1 2 3 4 5 6 7 8 9 0

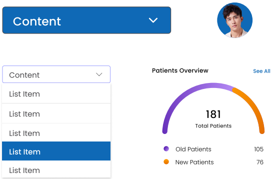
Example

Geometric sans serif typeface gives a breath of freshness,
modernity and gives the impression
of being user-friendly.

Icons



UI Elements



Content

Content

List Item

List Item

List Item

List Item

List Item

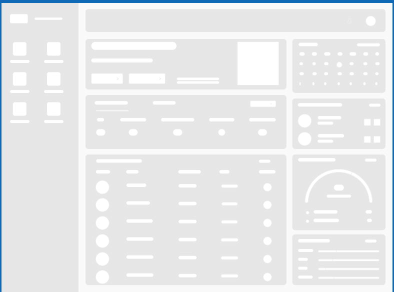
Patients Overview See All

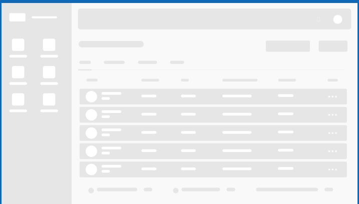
181
Total Patients

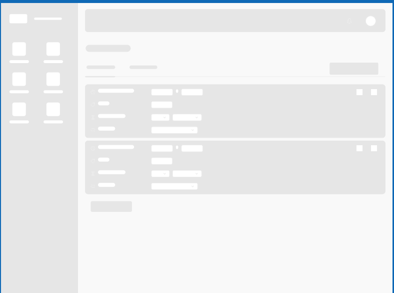
● Old Patients	105
● New Patients	76

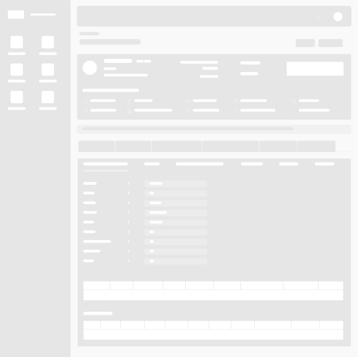
Design

Low Fidelity



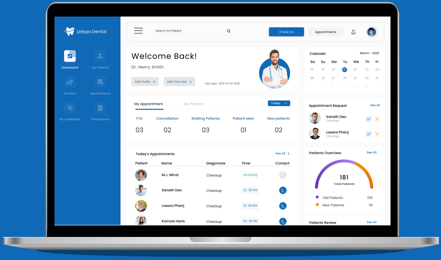






Final Designs

Dashboard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Edit Profile
Edit Time Slot
My Appointment
My Patients
Today
See All
See All
See All
See All

Welcome Back!

Dr. Henry Smith

Last Login: 2022-04-20 15:02

My Appointment

FTA	Cancellation	Waiting Patients	Patient seen	New patients
03	02	03	01	02

Today's Appointments

Patient	Name	Diagonosis	Time	Contact
	M.J. Mical	Checkup	On Going	
	Sanath Deo	Checkup	12 : 30 PM	☎
	Loeara Phanj	Checkup	01 : 00 PM	☎
	Komola Haris	Checkup	02 : 30 PM	☎

Calander March - 2022

Sa	Su	Mo	Tu	We	Th	Fr
18	19	20	21	22	23	24
25	26	27	28	29	30	31
1	2	3	4	5	6	7

Appointment Request

Sanath Deo
Checkup ✓ ✕

Loeara Phanj
Checkup ✓ ✕

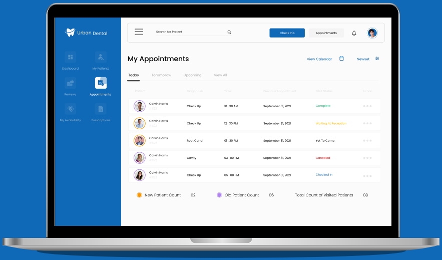
Patients Overview

181
Total Patients

● Old Patients 105
● New Patients 76

Patients Review

Appointments

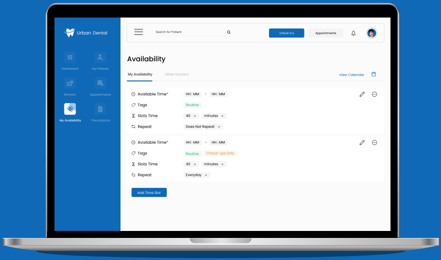

Urban Dental
Dashboard
My Patients
Reviews
Appointments
My Availability
Prescriptions
Search for Patient
Check In's
Appointments
View Calendar
Newset
Today
Tommorow
Upcoming
View All

My Appointments

Patient	Diagnosis	Time	Previous Appointment	Visit Status	Action
Calvin Harris #1122	Check Up	10 : 30 AM	September 31, 2021	Complete	•••
Calvin Harris #1122	Check Up	12 : 30 PM	September 31, 2021	Waiting At Reception	•••
Calvin Harris #1122	Root Canal	01 : 30 PM	September 31, 2021	Yet To Come	•••
Calvin Harris #1122	Cavity	03 : 00 PM	September 31, 2021	Canceled	•••
Calvin Harris #1122	Check Up	05 : 00 PM	September 31, 2021	Checked In	•••

● New Patient Count 02 ● Old Patient Count 06 Total Count of Visited Patients 08

My Availability

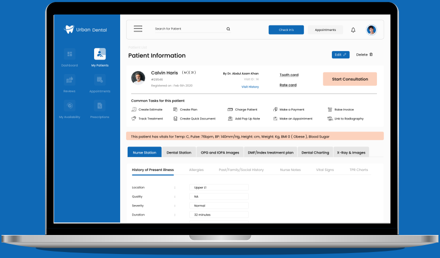

Urban Dental
Dashboard
My Patients
Reviews
Appointments
My Availability
Prescriptions
Search for Patient
Check In's
Appointments
My Availability
Other Doctors
View Calendar
Add Time Slot

Availability

Available Time* HH : MM - HH : MM
Tags Routine
Slots Time 40 minutes
Repeat Does Not Repeat

Available Time* HH : MM - HH : MM
Tags Routine Check-ups Only
Slots Time 40 minutes
Repeat Everyday

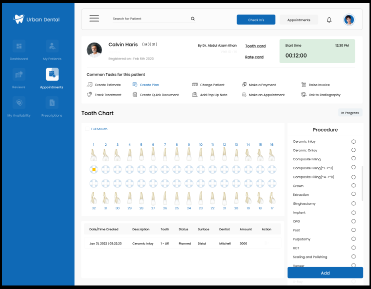
My patients


Urban Dental
Dashboard
My Patients
Reviews
Appointments
My Availability
Prescriptions
Search for Patient
Check In's
Appointments
Patient List
Patient Information
Edit
Delete
Calvin Haris
(M)(31)
#29546
Registered on : Feb 6th 2020
By Dr. Abdul Azam Khan
Visit ID : 14
Visit History
Tooth card
Rate card
Start Consultation
Common Tasks for this patient
Create Estimate
Create Plan
Charge Patient
Make a Payment
Raise Invoice
Track Treatment
Create Quick Document
Add Pop Up Note
Make an Appointment
Link to Radiography
Nurse Station
Dental Station
OPG and IOPA Images
DMF/Index treatment plan
Dental Charting
X-Ray & Images
History of Present Illness
Allergies
Past/Family/Social History
Nurse Notes
Vital Signs
TPR Charts

This patient has vitals for Temp: C, Pulse: 76bpm, BP: 140mm/Hg, Height: cm, Weight: Kg, BMI 0 (Obese), Blood Sugar

Location : Upper L1
Quality : NA
Severity : Normal
Duration : 32 minutes

Other Screens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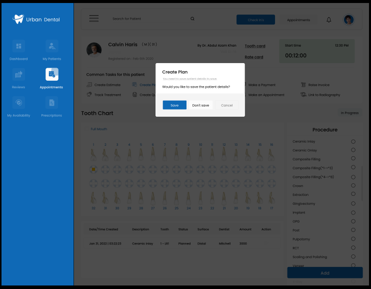
Calvin Haris
(M)(31)
Registered on : Feb 6th 2020
By Dr. Abdul Azam Khan
Tooth card
Rate card
Start time
12:30 PM
00:12:00
Common Tasks for this patient
Create Estimate
Create Plan
Charge Patient
Make a Payment
Raise Invoice
Track Treatment
Create Quick Document
Add Pop Up Note
Make an Appointment
Link to Radiography
Tooth Chart
In Progress
Full Mouth
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
32
31
30
29
28
27
26
25
24
23
22
21
20
19
18
17
Procedure
Ceramic Inlay
Ceramic Onlay
Composite Filling
Composite Filling(*1-=*3)
Composite Filling(*4-=*8)
Crown
Extraction
Gingivectomy
Implant
OPG
Post
Pulpotomy
RCT
Scaling and Polishing
Veneer
Add

Date/Time Created	Description	Tooth	Status	Surface	Dentist	Amount	Action
Jan 31, 2022 | 03:22:23	Ceramic Inlay	1 - LR1	Planned	Distal	Mitchell	3000	

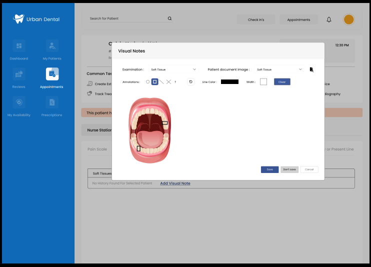

Urban Dental
Tooth Chart
Procedure
Add

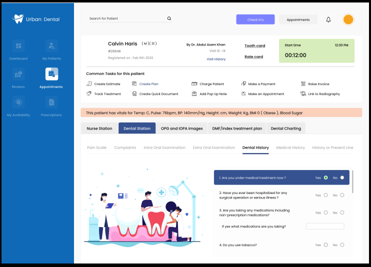
Create Plan

You need to save patient details to save.

Would you like to save the patient details?

Save
Don't save
Cancel


Urban Dental
Visual Notes
Examination :
Soft Tissue
Patient document image :
Soft Tissue
Annotations :
Line Color :
Width :
Clear
Save
Don't save
Cancel
Soft Tissues
No History Found For Selected Patient
Add Visual Note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Search for Patient
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Calvin Haris
(M)(31)
#29546
Registered on : Feb 6th 2020
By Dr. Abdul Azam Khan
Visit ID : 14
Visit History
Tooth card
Rate card
Start time
12:30 PM
00:12:00
Common Tasks for this patient
Nurse Station
Dental Station
OPG and IOPA Images
DMF/Index treatment plan
Dental Charting
Pain Scale
Complaints
Intra Oral Examination
Extra Oral Examination
Dental History
Medical History
History or Present Line

This patient has vitals for Temp: C, Pulse: 76bpm, BP: 140mm/Hg, Height: cm, Weight: Kg, BMI 0 (Obese), Blood Sugar

1. Are you under medical treatment now ? Yes ● No ○
2. Have you ever been hospitalized for any surgical operation or serious illness ? Yes ○ No ○
3. Are you taking any medications including non-prescription medications? Yes ○ No ○
If yes what medications are you taking?
4. Do you use tobacco? Yes ○ No ○

Test

“The final solution to answer the questions”

Empathize, define, and ideate have been carried out. So the next step is to make a prototype out of the mockup design I created. The prototyping stage was my chance to put everything I had learned so far to work.

In the last phase, I shared the prototype with my team and asked what could be improved and how it looked and felt to use. Two changes came out of it: a more compact appointments table that keeps only the useful information, and a visual tooth chart that makes treatment easier to explain to patients.